

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Second Semester 2024 - 2025
Comprehensive Examination (Closed book) – Part B
Electrical Sciences (EEE F111)

Time: 120 Minutes

Max Marks: 90

Date: 13 May 2025

Note: This question paper has 5 questions. Assume and clearly specify any missing data suitably. All parts of a question must be answered together.

Q1: For the circuit shown in Fig. 1, calculate the power dissipated in $4\ \Omega$ and $8\ \Omega$ resistors using superposition theorem. Also calculate the active power delivered by the voltage and current sources. [20]

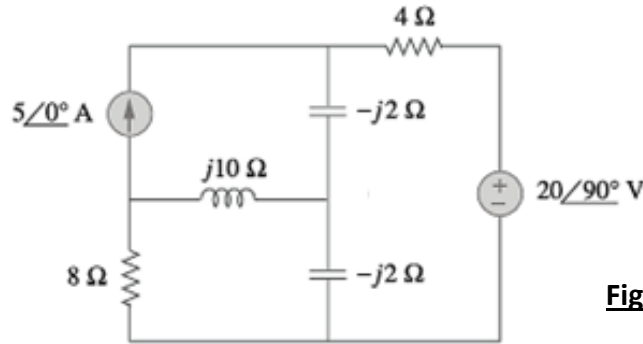


Fig. 1

Q2: Consider the ideal diode circuit shown in Fig. 2.

- (i) For the cases (a) and (b) given below, find the expression of the output voltage v_o as a function of the input voltage v_i , and also determine the corresponding ranges of the input voltage v_i . Zener breakdown voltage is 4 V.
 - (a) D_1 and D_2 are both in reverse bias region.
 - (b) D_1 is in the forward bias region and D_2 is in Zener breakdown region.
- (ii) Also, draw the output voltage waveform based on the range of input voltage conditions obtained in Part (i), for $v_i = 12 + \sin(\omega t)$ V.

[18]

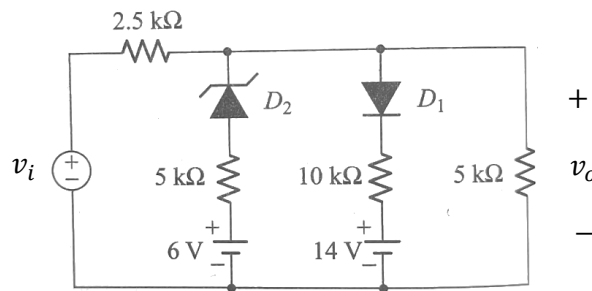


Fig. 2

Q3. For the BJT circuit shown in Fig. 3, determine the base and collector currents of each transistor. Also find the base, emitter, and collector voltages of each transistor. Assume both the transistors are working in the active region of operation. Given, $\beta_1 = 80$ (for Q_1) and $\beta_2 = 160$ (for Q_2). **[18]**

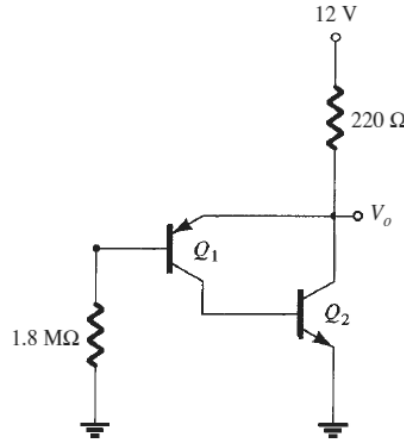


Fig. 3

Q4. For the MOSFET circuit shown in Fig. 4, the enhancement MOSFET (M_1) has $K = 0.25 \text{ mA/V}^2$ and $V_t = 1 \text{ V}$. The depletion MOSFET (M_2) has $I_{DSS} = 4 \text{ mA}$ and $V_p = -4 \text{ V}$. When, $V_{DD} = 12 \text{ V}$, $V_1 = 2 \text{ V}$, then M_1 is in the active region and M_2 is in the ohmic region. Find V_o and confirm the region of operation of both MOSFETs. The drain current equation in different regions for depletion type and enhancement type MOSFETs are given below. Use the appropriate conditions accordingly. **[16]**

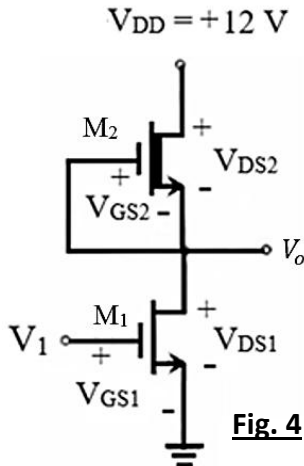


Fig. 4

Depletion type MOSFET

$$i_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

$$i_D = I_{DSS} \left[2 \left(1 - \frac{V_{GS}}{V_p} \right) \frac{V_{DS}}{-V_p} - \left(\frac{V_{DS}}{V_p} \right)^2 \right]$$

Enhancement type MOSFET

$$i_D = K(V_{GS} - V_t)^2$$

$$i_D = K[2(V_{GS} - V_t)v_{DS} - v_{DS}^2]$$

Q5. For the ideal transformer circuit shown in Fig. 5, find, (a) I_1 , (b) I_2 , (c) V_1 , (d) V_2 , and (e) V_o . **[18]**

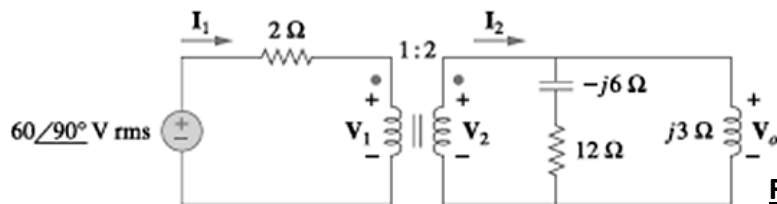


Fig. 5
